

Rwanda House Price Prediction

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Model: Multiple Linear Regression

1. Objective

The project develops a regression model to estimate house prices in Million Rwandan Francs from physical and location characteristics. The model is deployed as a Streamlit web application that returns an estimated price from user-entered house characteristics.

2. Data Preparation

The raw dataset contained 125 rows. 5 duplicate rows and 9 missing cells were identified during initial inspection. Exact duplicates were removed; missing numeric predictors were median-imputed, Neighborhood was mode-imputed, and rows without a target were removed (one row had no price). Negative numeric values were excluded; none were found. This left **119 cleaned rows** (125 - 5 duplicates - 1 missing target) with no missing values remaining.

IQR statistics were reviewed as an outlier screening step. On the price variable, the IQR upper bound was about 287 million RWF and 4 houses exceeded it (319, 322, 950 and 1,200 million RWF). The outliers were screened but **not removed**: the cleaned dataset still contains prices up to 1,200 million RWF and areas up to 1,200 m², so the model was trained and tested with them included.

3. Exploratory Analysis

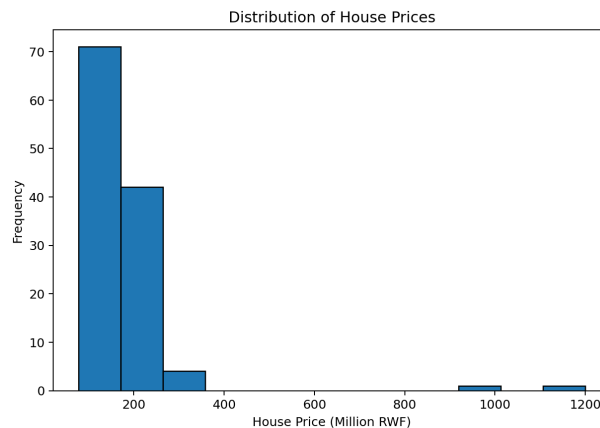


Figure 1. House-price distribution.

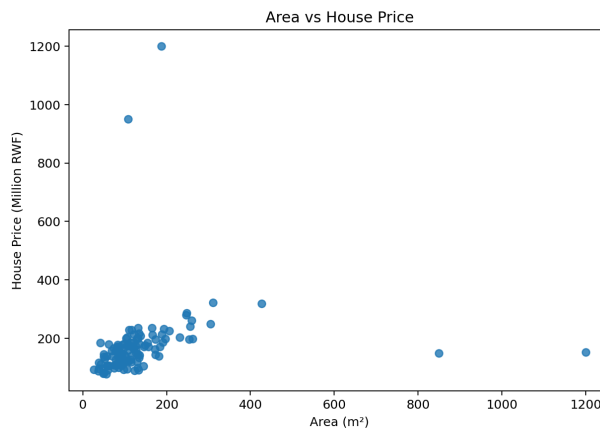


Figure 2. Area and price relationship.

4. Modelling

All required predictors except House_ID and the target were included. Neighborhood was one-hot encoded. An 80/20 train-test split with random_state=42 was used (95 training rows and 24 testing rows). Neighborhood has six categories (Gasabo, Huye, Kicukiro, Kigali City, Musanze, Nyarugenge); the first category, Gasabo, was dropped during encoding and acts as the baseline. The preprocessing and linear regression model were saved as one pipeline. The model uses the following seven predictors, which are also the inputs of the deployed application:

Predictor	Description	Type
Area_m2	Floor area of the house (m ²)	Numeric
Bedrooms	Number of bedrooms	Numeric
Bathrooms	Number of bathrooms	Numeric
House_Age_Years	Age of the house (years)	Numeric
Distance_to_City_km	Distance to the city centre (km)	Numeric
Parking_Spaces	Number of parking spaces	Numeric
Neighborhood	Location of the house (e.g. Gasabo)	Categorical (one-hot encoded)

Variable	Coefficient	Variable	Coefficient
Area_m2	0.1377	Neighborhood: Huye	-10.1253
Bedrooms	-7.4698	Neighborhood: Kicukiro	-5.6681
Bathrooms	43.9599	Neighborhood: Kigali City	31.2650
House_Age_Years	-0.9805	Neighborhood: Musanze	56.3815
Distance_to_City_km	0.6731	Neighborhood: Nyarugenge	20.6523
Parking_Spaces	18.9154		

Table 1. Fitted coefficients (intercept = 7.61 million RWF). Gasabo is the baseline neighborhood.

5. Results

Dataset	R ²	Adjusted R ²	MAE	RMSE
Training	0.2568	0.1583	38.50	81.05
Testing	-0.0188	-0.9526	83.28	213.80

Testing R² was -0.0188, with MAE 83.28 million RWF and RMSE 213.80 million RWF.

The test errors are dominated by a single house. One of the 24 test houses was priced at 1,200 million RWF and was predicted at roughly 200 million RWF (Figures 3 and 4). Excluding that house, the test MAE falls to 43.31 million RWF and the RMSE to 63.08 million RWF, although R² remains negative (-0.23). The model therefore does not explain price variation better than simply predicting the average price on this test set, and the negative adjusted R² reflects the small test sample (24 rows) relative to the 11 model terms.

6. Diagnostics

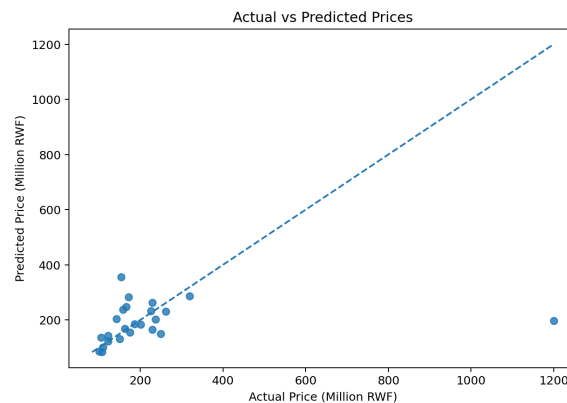


Figure 3. Actual versus predicted prices.

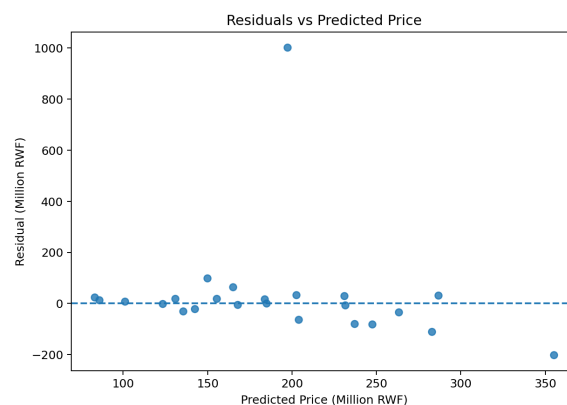


Figure 4. Residuals versus predicted prices.

7. Deployed Application

The saved pipeline is served through a Streamlit application titled "Rwanda House Price Prediction". The user enters the house characteristics (area, bedrooms, bathrooms, house age, distance to the city centre, parking spaces and neighborhood), and the application returns the estimated price in Million RWF. It also displays the submitted inputs as a table and provides a "Download Prediction CSV" button so the result can be saved.

Example prediction. For the input below, the application returned an estimated price of **130.06 Million RWF**.

Area_m2	Bedrooms	Bathrooms	House_Age_Years	Distance_to_City_km	Parking_Spaces	Neighborhood
150	3	2	5	5	2	Gasabo

Table 2. Example input and the resulting estimate (130.06 Million RWF).

8. Limitations and Conclusion

The dataset size (119 cleaned rows, only 24 in the test set) and the linear-model assumption limit generalisation. The extreme values (prices of 950 and 1,200 million RWF, and areas of 850 and 1,200 m² with ordinary prices) were retained, and a linear model is sensitive to them; this is the main reason for the weak test performance. Some coefficients, such as the negative effect of bedrooms, should be read with caution because predictors such as area, bedrooms and bathrooms are related to each other and the sample is small. Possible improvements include a larger dataset, a documented decision on the extreme values, and a log-transformed price or a non-linear model. The system should be interpreted as a price-estimation tool rather than a guaranteed market valuation. The final package contains the cleaned dataset, notebook, serialized pipeline, Streamlit app, dependencies, and report.